

BACK PAPER GEOMETRY EXAMINATION

Attempt any TEN questions. All questions carry equal marks. Total Marks 100

- (1) What is the reflection property of the parabola? Prove it.
- (2) What is a parallel projection of the plane? Give an example of an affine transformation which is not a parallel projection and show that any parallel projection is an affine transformation.
- (3) What is the fundamental theorem of affine geometry? Prove it.
- (4) State and prove Ceva's Theorem.
- (5) State and prove Menelaus' Theorem.
- (6) State and prove Desargues' Theorem.
- (7) State and prove Pappus' Theorem
- (8) State and prove Pascal's Theorem.
- (9) State and prove the fundamental theorem of inversive geometry.
- (10) State and prove the Coaxal Circles Theorem.
- (11) What is the hyperbolic distance between two points of the open unit disc? Prove that any hyperbolic circle is an Euclidean circle in the open unit disc, and vice versa.
- (12) Define a hyperbolic triangle and prove that the sum of the angles of such a triangle is less than π .
- (13) Prove that similar hyperbolic triangles are congruent.
- (14) Define a spherical triangle and prove that the sum of angles of a spherical triangle is greater than π .
- (15) Define the angular excess of a spherical triangle and prove that the area of a spherical triangle is equal to its angular excess.